§6.4.

P3. Proof:

$$(A^TCA)^T = A^TC^T(A^T)^T$$
$$= A^TCA$$
$$\Rightarrow A^TCA \text{ is symmetric}$$

P5. Solution:

$$A = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$$

$$\det(A-\lambda I) = \det\begin{bmatrix} -2-\lambda & 6 \\ 6 & 7-\lambda \end{bmatrix}$$
$$= (-2-\lambda)(7-\lambda) - 36$$
$$= \lambda^2 - 5\lambda - 50$$
$$= (\lambda-10)(\lambda+5) = 0$$
$$\Rightarrow \lambda_1 = 10, \quad \lambda_2 = -5$$

$\lambda_1 = 10$, solve $(A-10I)X_1 = 0$

$$\begin{bmatrix} -12 & 6 \\ 6 & -3 \end{bmatrix} X_1 = 0 \Rightarrow X_1 = c\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Let $\|X_1\|_2 = 1 \Rightarrow X_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$

$\lambda_2 = -5$, solve $(A+5I)X_2 = 0$

$$\begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} X_2 = 0 \Rightarrow X_2 = c\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Let $\|X_2\|_2 = 1 \Rightarrow X_2 = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}$

$$\Rightarrow Q = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix}$$

P11. solution:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(A-\lambda I) = \det\begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}$$
$$= \lambda^2 - 6\lambda + 8$$
$$= (\lambda-4)(\lambda-2) = 0$$
$$\Rightarrow \lambda_1 = 4, \quad \lambda_2 = 2$$

$\lambda_1 = 4$, solve $(A-4I)X_1 = 0$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} X_1 = 0 \Rightarrow X_1 = c\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$\Rightarrow X_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \text{ with } \|X_1\|_2 = 1$$

$\lambda_2 = 2$. solve $(A-2I)X_2 = 0$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} X_2 = 0 \Rightarrow X_2 = c\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$
$$\Rightarrow X_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \text{ with } \|X_2\|_2 = 0$$

$$\Rightarrow Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 4 & \\ & 2 \end{bmatrix}$$

$$\Rightarrow A = 4\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} - 2\begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

For $B = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$. we can find

$$Q = \begin{bmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 25 & \\ & 0 \end{bmatrix}$$
$$\Rightarrow A = 25\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\begin{bmatrix} 3/5 & 4/5 \end{bmatrix}$$

P14. <u>orthonormal</u>

$$\lambda = \pm i$$

P21. Solution:

(a) False (b) True

(c) True (d) False.

P23. Solution: <u>Flaw</u> - An eigenvector shared by A and A^T may not correspond to the same eigenvalue.

P26. Solution: $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

orthogonal $\Rightarrow ab + bc = 0$
$$\Rightarrow b(a+c) = 0$$

$$\Rightarrow A = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \text{ or } A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$
$$\Rightarrow \lambda = a \text{ or } \lambda = c \qquad \Rightarrow \lambda = \pm\sqrt{a^2+b^2}$$

§6.5

P2. Solution:

i) $A=\begin{bmatrix}1 & b\\ b & 9\end{bmatrix}$

$1>0$ & $9-b^2>0$

$\Rightarrow -3<b<3$.

LDL^T:

$A=\begin{bmatrix}1 & 0\\ b & 1\end{bmatrix}\begin{bmatrix}1 & \\ & 9-b^2\end{bmatrix}\begin{bmatrix}1 & b\\ 0 & 1\end{bmatrix}$

ii) $A=\begin{bmatrix}2 & 4\\ 4 & c\end{bmatrix}$

$2>0$ & $2c-4^2>0$

$\Rightarrow c>8$

LDL^T:

$A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}2 & \\ & c-8\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$

P3: Solution:

i) $A=\begin{bmatrix}1 & 2\\ 2 & 9\end{bmatrix}$

$f(x,y)=x^2+4xy+9y^2$

$=(x+2y)^2+5y^2$

ii) $A=\begin{bmatrix}1 & 3\\ 3 & 9\end{bmatrix}$

$f(x,y)=x^2+6xy+9y^2$

$=(x+3y)^2$

P6. solution:

$x^TA^TAx=(Ax)^T(Ax)$

$=y^Ty \quad (y=Ax)$

>0 if $y\neq 0$

$y=Ax\neq 0 \Rightarrow x\neq 0$ since A's columns are linearly independent.

$\Rightarrow x^T(A^TA)x>0$ except $x=0$.

$\Rightarrow A^TA$ is positive definite.

P8. Solution:

$f(x,y)=3(x+2y)^2+4y^2$

$=3x^2+12xy+16y^2$

$=[x\ y]\begin{bmatrix}3 & 6\\ 6 & 16\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}$

P13. Solution:

i) $A=\begin{bmatrix}c & 1 & 1\\ 1 & c & 1\\ 1 & 1 & c\end{bmatrix}$

$\det[c]=\underline{c>0}$

$\det\begin{bmatrix}c & 1\\ 1 & c\end{bmatrix}=c^2-1>0 \Rightarrow \underline{c>1}$ or $\underline{c<-1}$

$\det\begin{bmatrix}c & 1 & 1\\ 1 & c & 1\\ 1 & 1 & c\end{bmatrix}=(c-1)^2(c+2)>0 \Rightarrow \underline{c>-2 \text{ and } c\neq 1}$

$\Rightarrow c>1$

ii) $B=\begin{bmatrix}1 & 2 & 3\\ 2 & d & 4\\ 3 & 4 & 5\end{bmatrix}$

$\det[1]=1>0$

$\det\begin{bmatrix}1 & 2\\ 2 & d\end{bmatrix}=d-4>0 \rightarrow d>4$

$\det\begin{bmatrix}1 & 2 & 3\\ 2 & d & 4\\ 3 & 4 & 5\end{bmatrix}=-4d+12>0 \Rightarrow d<3$

$\Rightarrow B$ is not positive definite for any d.

P22. Solution:

i) $A=\begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix}$

we can find

$Q=\begin{bmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{bmatrix}$, $\Lambda=\begin{bmatrix}9 & \\ & 1\end{bmatrix}$ $(A=Q\Lambda Q^T)$

$\Rightarrow \Lambda^{\frac{1}{2}}=\begin{bmatrix}3 & \\ & 1\end{bmatrix} \Rightarrow$

$R=Q\Lambda^{\frac{1}{2}}Q=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$, $(R^2=A)$

ii) $A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$

We can find

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 16 & \\ & 4 \end{bmatrix}$$

$$\Rightarrow R = Q\Lambda^{\frac{1}{2}}Q = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \quad (R^2 = A)$$

P25. Solution:

$A = CC^T \to$ Cholesky factorization.

i) $C = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \Rightarrow A = CC^T = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$

ii) $A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 4 & \\ & 9 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\left(\begin{bmatrix} 2 & \\ & 3 \end{bmatrix}\begin{bmatrix} 2 & \\ & 3 \end{bmatrix}\right)\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & \\ & 3 \end{bmatrix}\right)\left(\begin{bmatrix} 2 & \\ & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\right)$$

$$= \left(\begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}\right)\left(\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}\right)$$

$$\Rightarrow C = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

P29. Solution:

i) $f_1(x,y) = \frac{1}{4}x^4 + x^2y + y^2 = (\frac{1}{2}x^2 + y)^2$

$$\begin{bmatrix} \frac{\partial f_1}{\partial x} \\ \frac{\partial f_1}{\partial y} \end{bmatrix} = \begin{bmatrix} x^3 + 2xy \\ x^2 + 2y \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 3x^2 & 2x \\ 2x & 2 \end{bmatrix}$$

$\det[3x^2] = 3x^2 > 0 \Rightarrow x \neq 0$

$\det A_1 = 2x^2 > 0 \Rightarrow x \neq 0$

$\Rightarrow A_1$ is $\begin{cases} \text{positive definite when } x \neq 0 \\ \text{positive semidefinite when } x = 0. \end{cases}$

$\Rightarrow f_1(x,y)$ is convex

$\nabla f_1 = 0 \Rightarrow \begin{cases} x^3 + 2xy = 0 \\ x^2 + 2y = 0 \end{cases}$

$\Rightarrow x^2 + 2y = 0.$

$\Rightarrow f_1(x,y) = 0$ are minimum on the curve $x^2 + 2y = 0$

ii) $f_2(x,y) = x^3 + xy - x$

$$\begin{bmatrix} \frac{\partial f_2}{\partial x} \\ \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 3x^2 + y - 1 \\ x \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 6x & 1 \\ 1 & 0 \end{bmatrix}$$

A_2 is clear indefinite for any (x, y) since $\det A_2 < 0$

$\nabla f_2 = 0 \Rightarrow \begin{cases} 3x^2 + y - 1 = 0 \\ x = 0 \end{cases} \Rightarrow \begin{matrix} x = 0 \\ y = 1 \end{matrix}$

$\Rightarrow (0, 1)$ is a saddle point of $f_2(x,y)$ with $f_2(0,1) = 0.$